Table 8.2: Conversion of Hexadecimal into Equivalent Binary Number	
Hexadecimal Number	**Equivalent Binary Number**
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Example: Conversion of a hexadecimal number 7B3 into its equivalent binary number.

7 *B* 3

0111 1011 0011

Hence, $7B3_{16} = 011110110011_{2}$

Note: Delete the left most zeros.

8.4.10 Binary to Hexadecimal Conversion

In this conversion, the binary number is arranged in group of 4 bits. Suppose the binary number grouping is not completed with the 4 digits, sufficient numbers of zero are added to the left most side of the binary number.

Example: Conversion of a binary number 110110101011100 into its equivalent hexadecimal number.

0110 1101 0101 1100

6 *D* 5 *C*

Hence, $110110101011100_{2} = 6D5C_{16}$

8.4.11 Decimal to BCD Conversion

In this method, each decimal digit is converted into its equivalent 4 digits binary form (BCD).

Example: Conversion of a decimal number 892 to its equivalent BCD number.

8 9 2

1000 1001 0010

Hence, $892_{10} = 100010010010_{BCD}$

8.4.12 BCD to Decimal Conversion

In this method, each BCD number grouped in the form of 4 digit binary pattern is converted into its equivalent decimal number.

Example: Convert a BCD number100100111000 to its equivalent decimal number.

1001 0011 1000

9 3 8

Hence, $100100111000_{BCD} = 938_{10}$

8.5 BINARY ADDITION AND SUBTRACTION

A logic circuit can be used to perform arithmetic functions like addition, subtraction, multiplication, division etc. For performing these operations, compliment method of number patterns are used. First, we will see the complement methods in order to understand the basic arithmetic operations.

8.5.1 One's Complement Method

In one's complement method, each binary bit of the number is changed from 0 to 1 or 1 to 0 depending on the existing bit value.

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